

Recitation 5 - Taxes

Taken from Fall 2023 by Cristine von Dessauer

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1 Refundable Taxes

Person A spends her income I between coffee C with price p_C and all other goods A with price p_A to maximize her utility $U = \frac{1}{2}\ln(C) + \frac{1}{2}\ln(A)$. Her marshallian demand for coffee is $d_C = \frac{I}{2p_C}$ and for all other goods is $d_A = \frac{I}{2p_A}$. Let $I = 10$, $p_C = 2$, $p_A = 1$

The government wants to implement a tax $\tau = \$0.5$ on coffee consumption and is considering two different options to refund said tax:

(a) Refund based on the pre-tax coffee quantity ($R = \tau \times C_{preTax}$).

(b) Refund based on the post-tax coffee quantity ($R = \tau \times C_{postTax}$).

1. Calculate Person A's demand for C and A , and her utility in absence of the tax.

Solution:

$$C^* = 2.5, A^* = 5, U_{preTax} = 1.262$$

2. Calculate Person A's demand for C and A , and her utility after the tax (without refund).

Solution:

$$C^* = \frac{10}{2*(2+0.5)} = 2, A^* = 5, U_{postTax} = 1.15$$

3. Calculate Person A's demand for C and A , and her utility after the tax and with refund (a).

Solution:

Without the tax Person A consumed $C = 2.5$ so the refund should be $R = 0.5 \times 2.5 = 1.25$ such that her total income is now $I_a = 11.25$. With this new income: $C^ = \frac{11.25}{2*(2.5)} = 2.25, A^* = 5.625, U_{refundA} = 1.269 > U_{preTax}$. Note that her previous bundle $C = 2.5$ and $A = 5$ is still purchasable but she chooses another one that gives her more utility.*

4. Calculate Person A's demand for C and A , and her utility after the tax and with refund (b).

Solution:

With the tax Person A consumes $C = 2$ so the refund should be $R = 0.5 \times 2 = 1$ such that her total income is now $I_b = 11$. With this new income: $C^ = \frac{11}{2*(2.5)} = 2.2, A^* = 5.5, U_{refundB} = 1.246 < U_{preTax}$. Note that her previous bundle $C = 2.5$ and $A = 5$ is no longer purchasable so she chooses another one that gives her less utility.*

2 Tax Incidence

The U.S. government is thinking about implementing a tax $\tau = 10$ on consumption on one of the two following markets:

Market 1: with aggregate supply $S_1 = 45P_1$ and aggregate demand $D_1 = 1000 - 5P_1$.

Market 2: with aggregate supply $S_2 = 5P_2$ and aggregate demand $D_2 = 9000 - 45P_2$.

The government wants to choose the market in which consumer incidence is the smallest.

1. Calculate the equilibrium (quantity and price) for both markets in the absence of the tax.

Solution:

Market 1: $P_1 = 20$, $Q_1 = 900$

Market 2: $P_2 = 180$, $Q_2 = 900$

2. Calculate the equilibrium prices (paid by consumers and received by producers) and quantities for both markets when a tax $\tau = 10$ on consumption is implemented.

Solution:

Market 1: $P_1^P = 19$, $P_1^C = 29$, $Q_1 = 855$

Market 2: $P_2^P = 171$, $P_2^C = 181$, $Q_2 = 855$

3. In which of the two markets did the tax have the smallest consumer incidence? Which market does the U.S. government choose?

Solution:

Market 1: consumer incidence = 9

Market 2: consumer incidence = 1

The U.S. government chooses market 2.

4. Suppose instead that aggregate supplies are such that for market 1, $p_1 = 20$ (perfectly elastic) and for market 2, $Q_2 = 900$ (perfectly inelastic). Answer questions 1-3 for this new set of supplies and demands.

Solution:

1. In absence of the tax:

Market 1: $P_1 = 20$, $Q_1 = 900$

Market 2: $P_2 = 180$, $Q_2 = 900$

2. In presence of the $\tau = 10$ tax:

Market 1: $P_1^P = 20$, $P_1^C = 30$, $Q_1 = 900$

Market 2: $P_2^P = 170$, $P_2^C = 180$, $Q_2 = 900$

3. In this case consumers absorb all of the tax incidence in market 1 (consumer incidence = 10) and absorb nothing of the tax incidence in market 2 (consumer incidence = 0). The U.S. government chooses market 2.